CHE 201 MidTerm Exam 2

April 13th, 2020

NAME: ________________________

Please fill in answers in the space supplied and <u>show all your work</u> to receive full credit. Useful Tables at the end of the Exam.

1. (25 points)
   A) State whether the following statements are **True of False (3 pts each)**

_______ The change in entropy of a closed system is the same for every process between two specified end states.

_______ When a substance undergoes a *throttling process* through a valve, the specific enthalpies of the substance at the valve inlet and valve exit are equal.

_______ A *nozzle* is a flow passage of varying cross-sectional area in which the velocity and the pressure of a gas or liquid increases in the direction of flow.

_______ The change in entropy (ΔS) for a system cannot be less than zero.

_______ The maximum coefficient of performance of *any* refrigeration cycle operating between cold and hot reservoirs at 100°C and 200°C, respectively, is closely 1.

_______ Every process consistent with the conservation of energy and conservation of mass principles can not occur in nature.

________ A thermodynamic process cannot convert heat completely into work.

B) (4 pts) The Reynolds number, Re, is a dimensionless number that measures the ratio of inertial forces to viscous forces in fluid flow;

$$Re = \frac{QD}{vA}$$

In this equation, Q is the volumetric flow rate, D is diameter of the pipe, A is cross-sectional area of the pipe, and v is the kinematic viscosity. What are the dimensions of v? **Show all your work for full credit.**

NAME: ______________________

CHE 201 MidTerm Exam 2 April 13th, 2020

2. (40 pts) It is recommended to cool your hot food to room temperature before putting into the fridge. This will save energy. A person is cooking large pan of dish three times a week with 4 kg of average mass of the pan and cooking. The average temperature of the kitchen is 25°C and the average temperature of the food is 90°C. The fridge is kept at 4°C, and the average specific heat of the food and pan can be approximated to 3.5kJ/kg. °C. And coefficient of performance for refrigerator is1.7.

a. (10 pts) Draw a schematic showing the refrigeration cycle, the thermal reservoirs, T_H, T_C, $\dot{W}_{cycle}$, $\dot{Q}_C$ and $\dot{Q}_H$. Show the direction of the energy transfer rates with arrows.

b. (5 points) Determine the amount of energy that refrigeration receives in a year to cool this dish with and without cooking to room temperature before putting into the fridge.
c. (5 points) Determine the amount of energy that refrigeration rejects in a year to cool this dish with and without cooking to room temperature before putting into the fridge.
d. (5 points) Determine the maximum coefficient of performance
e. (10 points) Determine how much this person will save a year by waiting for the food to cool to room temperature before putting into the fridge(12 cents/ kWh of electricity)
f. (5 points) Determine the entropy production, σ and state if the cycle is possible, impossible or reversible. Support your answer for full credit.

NAME: ________________________

CHE 201 MidTerm Exam 2 April 13^{th}, 2020

NAME: ________________________

CHE 201 MidTerm Exam 2 April 13th, 2020

3. (35 pts) A cogeneration system operates at steady state. Water vapor enters the system at 30 bar, 400^0C, velocity of 40m/s and mass flow rate of 2kg/s. Power is developed by the system at a rate of 2MW. There are two outlets for the system: process steam and hot water. Process steam leaves as saturated vapor at 1 bar, 100m/s, and hot water leaved as saturated liquid at 1 bar, 90m/s. Mass flow rates of process steam and hot water are equal. Neglect any changes in potential energy.

a) (5 pts) Draw a schematic of the system and label the system boundary.

b) (5 pts) State the engineering model

c) (5 pts) Write the simplified energy rate balance

d)(10pts) Calculate the rate of heat transfer, in MW, between the system and its surroundings.

e) (5pts) Calculate the volumetric flow rate at the inlet.

g) (5 pts) Sketch the process on a P-ν diagram. (mark specific volume for each state)

NAME: __________________________

CHE 201 MidTerm Exam 2 April 13^{th}, 2020